

MANET - I

Module 40

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Video: https://www.youtube.com/watch?v=7CR1_FBxJ6o

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0:02 [Music]

0:17 Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture

0:26 we will study about mobile ad hoc network first of all please see the contents which I shall

0:33 cover after covering brief introduction about Manet I shall take up characteristics of Manet which will be

0:41 followed by challenges of Manet networks in last I shall take up applications of

0:47 mobile ad hoc network so now let us start like traditional wir networks

0:55 wireless networks are formed by routers and hosts in a wireless network Network the routers are responsible for

1:02 forwarding data packets in the network and hosts may be sources or syns of data flows the fundamental difference between

1:11 wired and wireless networks is the way that the network components communicate a wi Network relies on

1:19 physical cables to transfer data in a wireless network the communication between different network components can

1:26 be either wired or Wireless since wireless communication does not have the constraint of physical

1:33 cables it allows a certain freedom for hosts and routers in a wireless network to

1:40 move this is one of the advantages of a wireless network according to the mobility of

1:47 hosts and routers there are three different types of wireless networks fixed wireless

1:54 network fixed hosts and routers use wireless channels to communicate at with each other and hence form a fixed

2:03 wireless network an example is a wireless network formed by fixed network devices using directed

2:11 antennas according to the figure two computers are communicating VI antenna
signals through this they can send or
2:19 receive the data wireless network with fixed access points mobile hosts use wireless
2:28 channels to communicate with fixed access points which may act as routers for
those mobile hosts and to form a
2:36 mobile network with fixed access points an example is the number of mobile laptop
users in a building that
2:44 access fixed access points according to the figure two or more laptops are sending
or receiving
2:53 the data via access points mobile ad hoc network a Manet is formed by mobile
3:01 3 minutes, 1 second hosts some of these mobile hosts are willing to forward
packets for Neighbors examples include vehicle to vehicle and
3:10 ship to ship networks that communicate with each other by relying on peer-to-peer
routings according to figure All Ships
3:19 create a network without any centralized Administration mechanism they can easily
establish the link with any node within
3:28 the network it is an infrastructure less IP based network of mobile and wireless
machine
3:35 nodes connected with radio the nodes of a Manet do not have centralized
Administration mechanism it is known for
3:44 its route able Network properties which each node act as a router to forward the
traffic to other specific node in the
3:54 network according to figure mobiles and laptops creates a network without any
centralized Administration in the
4:02 simplest scenarios nodes may be able to communicate directly with each other for
example when they are within Wireless
4:10 transmission range of each other however ad hoc networks must also support
communication between nodes that are
4:17 only indirectly connected by a series of Wireless hops through other nodes
according to figure to establish
4:25 communication between nodes A and C the network must enlist the aid of node B
to
4:32 relay packets between them these circles indicate the nominal range of each nodes
radio
4:39 transceiver nodes A and C are not in direct transmission range of each other
4:46 since a circle does not cover C characteristics of mobile ad hoc
4:54 networks as compared to other wired or infrastructure based wireless networks
Manet has the following characteristics
5:02 autonomous terminal distributed operation multi hop routing lightweight

5:09 terminals Dynamic topology self organization multi hoping resource
5:18 conversation limited capacity and scalability autonomous terminal in
5:26 manate each mobile terminal is an autonomous node which may function as a host
or a router in other words beside
5:35 the basic processing ability as a host the mobile nodes can also perform switching
functions as a router so
5:42 usually endpoints and switches are indistinguishable in Manet distributed
operation since there is no background
5:50 Network for the central control of the network operations the control and
management of the network is distributed
5:58 among the terminals the nodes involved in a Manet should collaborate Among Us
themselves and each node acts as a relay
6:07 as needed to implement functions like security and routing multi hope routing
6:14 in multihop routing ad hoc routing algorithms can be single hope or multi hope
they are based on different link
6:23 layer attributes and routing protocols single hope Manet is simpler than multi hope
in terms of structure and
6:32 implementation while transferring data packets from a source to its destination out
of the direct Wireless transmission
6:40 range the packets should be forwarded by intermediate nodes lightweight
terminals
6:47 in most of the cases the Manet nodes are mobile devices with less CPU processing
capability small memory size and low
6:56 power storage such devices need optimized algorithms and mechanisms that
implement the Computing and communicating
7:05 functions Dynamic topology all nodes of Manet are free to move causing Network
7:12 topology change rapidly at unpredictable times links between nodes are expected
to break much more frequently than with
7:21 wired and infrastructure based wireless networks self organization due to the lack
of infrastructure or Central
7:29 Administration nodes should be able to form themselves into a network multi
hoping in a Manet nodes
7:39 use a wireless channel to transmit data and due to the limited number of nodes
neighbors intermediate nodes are used to
7:47 relay the packets resource conservation in manner the nodes are limited in both
energy Supply and processing
7:56 power power conservation becomes a very important factor to be considered when
designing a network therefore optimizing
8:05 all operations May minimize the energy consumption limited security mobile ad hoc
networks are more prone to security

8:13 threats than wi networks or infrastructure based wireless networks because of their unique

8:21 characteristics each mobile node in an ad hoc network and function as a router or packet forwarded for other nodes

8:30 both legitimate users and malicious attackers can access the wireless Channel and there is no well place where

8:38 excess control mechanisms can be deployed scalability in some applications mobile

8:45 ad hoc networks May grow up to several thousand nodes mobile ad hoc networks suffer from scalability problems in

8:53 Channel capacity because Channel capacities are very limited and maximum use of Channel capacity can be reached

9:01 9 minutes, 1 second faster due to the multi- hopping nature of mobile ad hoc networks their scalability is related to the routing

9:08 protocols they employ low bandwidth these networks have lower capacity and shorter transmission range

9:16 than fixed infrastructure networks the throughput of wireless communication is lesser than wir communication because of

9:25 the effect of multiple excess fading noise and and interference conditions

9:31 challenges of mobile ad hoc networks in last few years mobile ad hoc networking has been a popular field of

9:40 communication almost every aspect of the network has been explored in one way or other at different level of problem the

9:49 topics that need to be resolved are as follows scalability routing quality of service

9:58 client server model model shift security energy conservation node cooperation

10:07 interoperation scalability most of The Visionaries depicting applications which are anticipated to benefit from the adhoc

10:16 technology take scalability as granted imagine for example the vision of

10:22 universal Computing where networks can be of any size however it is unclear how

10:30 such large networks can actually grow adhoc networks suffer by Nature from the

10:36 scalability problems in capacity routing routing in Wireless ad hoc networks is non-trivial due to the highly Dynamic

10:46 environment an ad hoc network is a collection of Wireless Mobile nodes dynamically forming a temporary Network

10:53 without the use of any pre-existing Network infrastructure or centralized Administration

11:00 in a typical ad hoc network mobile nodes come together for a period of time to exchange information while exchanging

11:08 information the nodes may continue to move and so the network must be prepared to adapt continually to establish Roots

11:16 among themselves without any outside support quality of service the
11:23 heterogeneity of existing internet applications has challenged Network designers
who have built the network to
11:31 provide best effort service only Voice live video and file transfer are just a
11:38 few applications having vast and diverse requirements quality of service that is
11:45 Q aware Solutions are being developed to meet the emerging requirements of these
11:51 applications qos has to be guaranteed by the network to provide certain
performance for a given flow or or a
11:59 collection of flows in terms of Qs parameters such as delay Jitter
12:06 bandwidth packet loss probability and so on despite the current research efforts
12:12 in Qs areas Qs in ad hoc networks is still an unexplored area client server
12:20 model in the internet a network client is typically configured to use a server as its
partner for Network transactions the
12:29 these servers can be found automatically or by a static configuration in ad hoc
networks however the network structure
12:37 cannot be defined by collecting IP addresses into subnets there may not be servers
but the demand for basic
12:45 Services still exists address allocation name resolution authentication and the
12:52 service location itself are just examples of basic Services which are needed but
their location in the network
12:59 is unknown and dynamically changes over time due to the infrastructure less
nature of these networks and nodes
13:08 Mobility a different addressing approach may be required in addition it is still not
clear who will be responsible for
13:16 managing various Network Services therefore while there have been vast research
initiatives in this area the
13:24 issue for shifting from the traditional client server model remains to be
appropriately addressed security a vital
13:34 issue that has to be addressed is the security in ad hoc networks applications like
military and confidential meetings
13:42 require High degree of security against enemies and active passive attackers ad
hoc networks are
13:50 particularly prone to malicious behavior lack of any centralized Network
management or certification Authority
13:58 makes these dynamically changing Wireless structures security is often considered
14:04 to be the major Road Block in commercial application interoperation when two
autonomous ad
14:12 hoc networks move into same area the interference with each other becomes
unavoidable and the networks would

14:21 recognize the situation and be merged however the issue of joining two networks is not trivial the networks may

14:28 be used using different synchronization or even different Mac or routing protocols security also becomes a major

14:36 concern can the networks adapt to the situation for example a military unit moving into an area covered by a sensor

14:45 Network could be such a situation moving unit would probably be using different routing protocol with location

14:53 information support while the sensor Network might be using a simple static router protocol so now we move on to the

15:02 applications of ad hoc networks mobile ad hoc networks are best suitable for situations where an

15:09 infrastructure is not available or to deploy one is not cost effective the following are some of the important

15:18 applications military applications mobile ad hoc networks satisfies several military needs such as Battlefield

15:26 survivability in such environment setting up of an infrastructure for communication between shoulders in

15:33 Battlefield could be impossible the wireless devices carried by shoulders can form a mobile ad hoc network to

15:41 support communication among them according to the figure shoulders tanks helicopter and other vehicles carry the

15:50 mobile devices for message passing process home office and educational applications ad hoc networks also have

15:59 applications in Home and Office environments the simplest and most direct application of ad hoc networks in

16:07 both homes and offices is the networking of laptops pdas and other WLAN enabl

16:15 devices in the absence of Wireless based station another home application that falls within the personal area network

16:23 that is pan class as wire replacement through Wireless links and as in Bluetooth all periphery devices can

16:32 connect to computer through wireless Bluetooth links eliminating the need for wide connections ad hoc networks can also

16:41 enable streaming of video and audio among Wireless nodes even in the absence of any base station for example students

16:50 in a classroom can use their laptops to obtain the latest class material from a professor's laptop as the class progresses

16:59 universities and campus settings virtual classrooms ad hoc Communications during meetings or lectures are some of the

17:07 educational applications of ad hoc networks a vehicular ad hoc network ret
vehicular ad hoc networks are

17:16 responsible for the communication between moving vehicles in a certain
environment a vehicle can communicate

17:23 with another vehicle directly which is called vehicle to vehicle that is v2v communic
ation or a vehicle can

17:31 communicate to an infrastructure such as roadside unit RSU known as vehicle to
infrastructure

17:39 v2i according to figure vanet scenario is creating VI v2v communication and v2i

17:46 communication between the vehicles on the road wireless sensor networks wireless
sensor networks can be considered as

17:55 special case of mobile ad hoc networks Manet with reduced or no Mobility initially
wsn was mainly

18:05 motivated by military applications on the civilian application domain of wireless
sensor networks such as

18:13 environmental and spaces monitoring disaster management smart home production
and Healthcare

18:21 Etc these wsn May consist of heterogeneous and mobile sensor nodes the Network
topology

18:29 may be as simple as start topology the scale and density of a network varies
depending on the

18:37 application according to the figure application of wireless sensor networks include
disaster management Network

18:45 management home networks Healthcare informatics and Military applications
sensor networks have

18:53 emerged as a promising tool for monitoring the physical world utilizing self
organization networks of battery

19:02 powered Wireless sensors that can sense process and communicate a sensor
network is a

19:09 network of many tiny disposable lower part devices called nodes which are
specially distributed in order to

19:17 perform an application oriented Global task these nodes form a network by

19:25 communicating with each other either directly or through other nodes one or more
nodes among them will

19:33 serve as sync that are capable of communicating with the user either directly or
through the existing wild

19:41 networks the primary component of the network is the sensor essential for
monitoring real world physical

19:49 conditions such as sound temperature humidity vibration pressure motion

19:55 pollutant Etc at different locations wireless mesh networks wireless mesh
20:02 networks can easily effectively and wirelessly connect entire cities using
20:09 inexpensive existing technology traditional networks rely on a small number of
wired access points or
20:16 wireless hotspots to connect users in a wireless mesh Network the network
connection is spread out among dozens or
20:25 even hundreds of wireless mesh nodes that talk talk to each other to share the
network connection across a large
20:33 area mesh nodes are small radio transmitters that function in the same way as
wireless router nodes use the
20:42 common Wi-Fi standards known as 802.11 a B and G to communicate
20:50 wirelessly with users and with each other it works on 2.4 GHz and 5 GHz
20:58 frequency bands depending on the physical layer used for example if i e
21:05 802.11a is used the speed can be up to 54 Mbps nodes are programmed with
21:13 software that tells them how to interact with a large Network information travels
across the network from point A to point
21:21 B by hopping wirelessly from one mesh node to the next the nodes automatically
21:28 choose the quickest and safest path using a process known as Dynamic routing
according to the figure there is
21:36 an internet Cloud which is the backbone of mesh routers mesh routers are
connected with
21:43 access points base stations and other network devices all network devices provide
the
21:50 internet access functionality to all connected users in the network disaster relief
operations
21:59 every year natural disasters like earthquake flood destroy people's lives around the
world as the importance of
22:08 the internet grows the loss of network connectivity during such disasters will be
more noticeable effect of The
22:16 Misfortune so it is important to find ways to enable the operations of networks even
when infrastructure
22:24 elements are disabled as a result of disasters personal area networks a personal
area
22:32 network that is pan is a computer network used for data transmission among
devices such as computers telephones and
22:41 personal digital assistants the idea of a personal area network is to create a
network that
22:48 consists of nodes which are associated with a single person these nodes may be
placed in a

- 22:55 person's clothes belt or carried in handle bags this type of network provides great flexibility for example
- 23:04 it allows you to send a document to the printer in the office upstairs while you are sitting in the couch with your
- 23:11 laptop using pan you can upload the photo from your cell phone to your desktop computer so friends this was all
- 23:21 about Manet hope the concepts explained in this lecture were understandable and helpful in the next next lecture of
- 23:29 Manet I will cover the routing protocols used in mobile ad hoc network hope to see you in the next
- 23:36 lecture till then goodbye thank you
- 23:41 [Music]